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Makefile

paper0.tex

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llm.tex

paper1.tex

paper0.tex

Code Visual Editing

1 \documentclass[sigconf,nonacm]{acmart}

2 \usepackage{booktabs}

3 \usepackage{tikz}

4 \usetikzlibrary{decorations.pathmorphing}

5 \settopmatter{printfolios=true}

6 \input{fig/litnums}

7

8 % review comments: \note{who}{what}; resolve

9 % make notes lists them all. acmart already

10 \newcommand{\note}[2]{\textcolor{red}

11 {\textbf{[1: #2]}}

12 % pause-and-review call-out boxes:

13 \begin{pause}{topic}...

14 \newcounter{pausebox}

15 \newsavebox{\pauseboxsave}

16 \newenvironment{pause}[1]{%

17 \refstepcounter{pausebox}\par\medskip\noinde

18 nt

19 \begin{lrbox}{\pauseboxsave}%

20 \begin{minipage}{\dimexpr\linewidth-9mm}%

21 \small\noindent

22 \textbf{[1: #2]}\& review

23 \thepausebox: #1}\[0.2em]}%

24 {\end{minipage}\end{lrbox}%

25 \noindent\begin{tikzpicture}

26 \node[inner sep=5pt, anchor=west] (b) at

27 (5mm,0)

28 {\usebox{\pauseboxsave}};

29 \fill[black!12]

30 ([xshift=-5mm]b.north west) rectangle

31 (b.south west);

32 \draw[decorate, decoration={random steps,

33 segment length=7pt,

34 amplitude=0.7pt}, line width=0.6pt]

35 ([xshift=-5mm]b.north west) -- (b.north

36 east) --

37 (b.south east) -- (b.south west) --

38 ([xshift=-5mm]b.south west) -- cycle;

39 \end{tikzpicture}\par\medskip}

40 % compact author block: one names line, one

41 affils line

42 \makeatletter

43 \renewcommand\@authorfont{\Large\sffamily}

44 \renewcommand\@affiliationfont{\normalsize}

45 \makeatother

46 \title{Follow Your Strengths: Writing Papers

47 with LLMs}

48 % note in title breaks headers; working

49 title, timm to confirm.

50 \author{Tim Menzies1,

Recompile

23

5)).

Given this, we take guidance from two sources. The first is Newell's advice to young scientists, retold by Shaw (who worked

¹"Slop" is low-quality machine-generated content produced at volume; Merriam-Webster made it the 2025 word of the year [6]. Its reach into science is measurable: one excess-vocabulary study of 14 million PubMed abstracts estimated that at least 10% of 2024 abstracts passed through an LLM [3]. Software developers report the same burden around AI-generated code and documentation [1]. The term still lacks an agreed definition, though its dimensions (e.g. coherence, relevance, information density) are being mapped [9].

- automatically generated critiques which, acting as ICSE's reviewer two, report the weaknesses of the current draft and propose a revision that avoids at least some of them;
- etc.

Hence our second problem also yields, since LLM prose need not be bland once the machine is told, precisely, what bland looks like. Before starting, two frequently asked questions deserve answers.

- Should we follow Newell's advice at all? If everyone focuses on their own tricks, each of us risks the guilt of the hammer that sees only nails. Our short reply is that strengths choose the tools while the field's open problems choose the targets; hence the hammer is aimed, not wandering. In our experience the method does not rut a career, for two reasons: its literature step keeps forcing contact with material outside prior experience, and its experiments force a critical evaluation of the tool base's own premises.
- Are we trying to automate science? No. We aim to reduce the effort scientists spend working together. The process of this paper is not a CI pipeline where resumes drop in at one end and papers roll off the other. Rather, it pauses repeatedly at numbered review points where humans receive succinct conclusions to inspect, dispute, and change; every change re-runs the downstream steps. In what follows, those pauses appear as boxed call-outs, each listing the questions a reviewer should ask and naming exactly what to edit so that the rest of the processing follows the reviewer, not the machine.

[claude: roadmap paragraph goes here once paper0's sections exist]

2 Setting Up to Write Together

The advice of the last section assumes a team, and teams arrive

- a map of the common structures of research papers;

²Here, "recent" might mean the last five years. However, for fast-moving fields such as LLMs, "recent" might be as little as 12 to 24 months.

³For newcomers to a field, we recommend a small variant of this question: for the people we can talk to in our local area, what are their recent best papers?

⁴We do not presume to say what "important issues" might mean to the reader. It could be many things: the area of most current commercial interest; a "hole", i.e. an area no one is looking at; a "mountain", i.e. an area everyone is working on; etc. For mountains, two questions can be insightful: (a) if everyone is doing something, is everyone doing that something wrong (in some way)? (b) if everyone is doing something, can we optimize that something (even a little)?

1

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Merges stay small because of two source-layout habits. Firstly, every sentence ends its line, so git diffs and merges by the sentence, and two edits collide only when they touch the same sentence. Secondly, prose lives in many small files (one per section) beneath a thin root of include statements, so writers claim a section, not the paper. Review comments follow the same philosophy: a comment is a macro in the source (who said it, what they said), listed by one make target and resolved by deleting it. Margin comments in web editors were rejected since they do not survive the trip through git.

The last ritual is scheduling. Before any shared writing session, one owner verifies that every participant can already write to something: doc links delivered, project invites accepted, push access proven by one trivial commit. Access problems solved during a session cost n people an hour; solved the day before, they cost one person a minute. The prompts the team uses are versioned in the same repository as the prose, so every rung (browser paste or agent invocation) runs the same words. (Aside: this paper was written with exactly this setup, one author plus one agent, trading pushes through the bridge described above.)

Table 1 makes the ladder concrete for this paper's own repository: what each co-author does once, and what they do daily.

3 Know Thyself

Know thyself. — inscribed on the Temple of Apollo at Delphi

Newell's advice cannot be followed until the strengths are named, and self-report names them badly. Hence, before diving deep into any literature, we first reverse engineer the team from the record of its own papers.

Our instrument is the repertory grid of Kelly's personal construct theory [2]. Kelly held that people make sense of a domain through bipolar distinctions (*constructs*) of their own devising. A grid is elicited by triads: pick three team members, ask which one stands